

Week 2 Assignment 2

Q1. A bank wants to predict whether a loan application will be approved or rejected.

Classification or Regression?

It is a Classification problem because the output belongs to one of two categories: Approved or Rejected.

Most Suitable Algorithm

Among Decision Tree, KNN, and Linear Regression, Decision Tree is the most suitable because it works well for classification tasks and provides easy-to-understand decision rules.

Q2. Explain the difference between a feature and a label using a real-world example.

A feature is an input variable used to make predictions.

A label is the output variable that the model predicts.

Example: House Price Prediction

Features:

- Number of rooms
- Area of the house
- Location

Label:

- House price
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Q3. A Decision Tree predicts that a customer will purchase a product. Explain how the model reaches this prediction.

A Decision Tree makes predictions by asking a sequence of questions based on the customer's characteristics.

For example:

- Is the customer's age greater than 30?
- Has the customer purchased before?
- Is the customer's income above ₹50,000?

Based on the answers, the model follows a path through the tree until it reaches a final prediction such as "Will Purchase".

Q4. Why is KNN considered a supervised learning algorithm? Can KNN perform classification if labels are not available?

KNN is considered a supervised learning algorithm because it learns from labeled training data.

No, KNN cannot perform classification if labels are not available because it requires known labels to determine the class of new data points.

Q5. Suppose a KNN model uses $K = 5$ and the nearest neighbors are:

Yes, Yes, No, Yes, No

Prediction: Yes

Reason: KNN uses majority voting. Since "Yes" appears 3 times and "No" appears 2 times, the model predicts "Yes".

Q6. Explain the equation of Linear Regression:

$$y = mx + b$$

Where:

- x = Independent variable (input)
- y = Dependent variable (output)
- m = Slope of the line
- b = Intercept (value of y when $x = 0$)

Example: Predicting salary based on years of experience.

Q7. What is correlation?

Correlation measures the strength and direction of the relationship between two variables.

+1

Perfect positive correlation. As one variable increases, the other also increases.

0

No correlation. There is no relationship between the variables.

-1

Perfect negative correlation. As one variable increases, the other decreases.

Q8. A disease detection model produced the following result:

Actual Condition: Disease Present

Prediction: Healthy

Type of Prediction: False Negative (FN)

Why is it dangerous?

This is dangerous because the patient actually has the disease, but the model predicts that the patient is healthy. This may delay treatment and lead to serious health consequences.

Q9. What is the purpose of a Confusion Matrix? How is it different from a Correlation Matrix?

Confusion Matrix

A Confusion Matrix is used to evaluate the performance of a classification model. It shows:

- True Positive (TP)
- True Negative (TN)
- False Positive (FP)
- False Negative (FN)

Correlation Matrix

A Correlation Matrix shows the relationship between numerical variables in a dataset.

Difference:

- Confusion Matrix evaluates classification performance.
 - Correlation Matrix measures relationships between variables.
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Q10. Compare the following algorithms in one or two lines each.

Decision Tree

A tree-based algorithm that makes decisions using a series of conditions and rules.

KNN

Classifies data based on the majority class of its nearest neighbors.

SVM

Finds the best boundary (hyperplane) that separates different classes.

Linear Regression

Predicts continuous numerical values using a linear relationship between variables.

Neural Network

Uses interconnected layers of artificial neurons to learn complex patterns from data.

